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LARYNGEAL REALISM AND THE PREHISTORY OF CELTIC[1]

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Abstract

This paper examines the proto-Celtic plosive system through the lens of Laryngeal Realism. Drawing upon phonetic data from contemporary Celtic languages and philological data from medieval Insular Celtic and ancient Continental Celtic languages, it concludes that the active Laryngeal feature in these languages is not [voice], but [spread glottis], and that this feature should be projected back to proto-Celtic. Such an analysis allows for a much more straightforward analysis of the evolution of the early Celtic plosive system, and, in particular, allows for a non-stipulative analysis of perhaps the best known of Celtic sound changes, the loss of proto-IE */p/, in simple aerodynamic terms. It is demonstrated, furthermore, that the loss of proto-IE */p/ cannot be explained by contact with pre-Basque or Iberian, but, instead, was, in all likelihood, a natural development.

1. Prelude

This paper examines prehistoric Celtic sound changes in the plosive system through the lens of Laryngeal Realism, an approach to phonology which 'adopts a more transparent relationship between distinctive features as phonological entities and the phonetic substance of those features' (definition as per Brown 2016: 397; the term 'Laryngeal Realism' originates with Honeybone 2005).

There are three Laryngeal features:

(1) Laryngeal

voice | spread glottis | constricted glottis

In languages that possess a binary contrast between plosives in which the feature [voice] is active, /p t k/ contrast with /b d ɡ/ (e.g. Spanish); in languages in which [spread glottis] is active, /pʰ tʰ kʰ/ contrast with /p t k/ (e.g. English);[2] in languages in which [constricted glottis] is active, /p' t' k'/ contrast with /p t k/ (e.g. St'át'imcets).[3]

[1] I should like to thank the editor, James Clackson, and the anonymous reviewers for their helpful comments. All usual disclaimers apply.

[2] Tape cutting experiments of words commencing with /s/ plus voiceless plosive make this contrast clear for English. Such sequences are taped, the friction of the /s/ removed, and subjects asked to identify the remaining word; for example, *~~s~~py*, *~~s~~ty*, and *~~s~~ky* were heard as *buy*, *dye*, and *guy* once the sibilant was subtracted in virtually all tokens (Lotz et al.1960; Reeds & Wang 1961).

[3] It has been claimed that Swedish (Ringen & Helgason 2004; Petrova et al. 2006; Helgason & Ringen 2008; Beckman et al. 2011) and Norwegian (Ringen & van Dommelen 2013) contrast [spread glottis] and [voice], but Salmons (forthcoming) demonstrates that [voice] is phonologically inert in these languages, and that any voicing present is for phonetic enhancement purposes (see Avery & Idsardi 2001; Keyser & Stevens 2006; Stevens & Keyser 2010; Hall 2011 on the deployment of non-contrastive gestures to render phonological contrasts more perceptually salient).

2. The active Laryngeal feature in contemporary Celtic languages

Grammars of Celtic languages, ancient, medieval, and contemporary, presume that [voice] is the active Laryngeal feature in the plosive system and that they contrast /p t k/ vs. /b d g/ (see, for example, Lambert 2003: 45 for Transalpine Celtic; Thurneysen 1946: 114–118 for Old Irish; Jackson 1953: 394–436 passim for the medieval Brittonic languages; Ó Siadhail 1989: 82 for contemporary Irish; Broderick 1984: 3–4; 1986: 3–13 for late spoken Manx;[4] Thorne 1993: 7 for contemporary Welsh; Sommerfelt 1978: 48–56 for contemporary Breton).[5]

But this is manifestly untrue for the contemporary Celtic languages:

(2) a. Welsh:
Jones (1984: 41) states that 'the series /p, t, k/ are aspirated and voiceless, while the series /b, d, g/ are unaspirated, and although frequently referred to as voiced, voicing is not a constant feature of their articulation. They are regularly unvoiced in word initial and final positions . . . and frequently so in a medial fully voiced environment. They may be partially voiced in all these environments, but fully voiced occasionally in medial fully voiced environments only'.

b. Breton:
Ternes (2011: 448) states that '[s]timmlose Plosive sind mittelstark aspiriert'; Bothorel 1982: 342 reports that 'la réalisation est toujours sonore' for voiced plosives in intervocalic position, but that they 'ne sont pas nécessairement sonores' in initial position and have a 'tendance à être réalisées sourdes'. In final position, 'la règle est à la neutralisation en faveur de la sourde'.[6]

c. Irish:
Ní Chasaide (1999: 113) states that '[t]he voiceless [plosive] series is strongly postaspirated and devoicing of the voiced [plosive] series is fairly widespread, especially in non-intervocalic contexts'.

That [spread glottis] is the active Laryngeal feature in contemporary Celtic languages is demonstrated by the positive values for Voice Onset Time (VOT) measurements (measured in milliseconds). We may compare the values for Welsh (Ball 1984: 15) with those for Scottish Gaelic (Nance & Stuart-Smith 2013: 137), which reveal that Welsh, generally presumed to be a [voice] language, is more aspirated at some places than Scottish Gaelic, a [spread glottis] language:[7]

(3)

	$/p^h/$	/p/	$/t^h/$	/t/	$/k^h/$	/k/
Welsh	62	15	82	33	97	32
Scottish Gaelic	71	17	73	23	89	33

Compare these measurements with those of a [voice] language such as Spanish (Lisker & Abramson 1964: 392), in which the negative values for the voiced plosives indicate that voicing commences well before the release of the occlusion:

[4] Broderick (1984: 4) notes that the coronal plosives in late spoken Manx tended towards being distinguished by aspiration.

[5] Scottish Gaelic, which is universally recognised to contrast $/p^h\ t^h\ k^h/$ vs. /p t k/, and, hence, is a [spread glottis] language, is an exception (cf. Borgstrøm 1940: 20–21, 130–131; 1941: 33–34, 91; Oftedal 1956: 98–99; MacAulay 1992: 230–231; Ladefoged et al. 1998: 4–12; Ternes 2006: 9–10; Gillies 2010: 232).

[6] Many dialects of Breton do appear to have a Laryngeal contrast via [voice], likely due to prolonged contact with French, a [voice] language (see Caramazza & Yeni-Komshian 1974; Fougeron & Smith 1993: 75).

[7] Welby et al. (2017: 139) provide VOT data for Irish at the labial and dorsal places ($/p^h/$ = 62, /p/ = 15, $/k^h/$ = 64, /k/ = 21), which indicate that it, like Welsh and Scottish Gaelic, is a [spread glottis] language (N.B. that they label /p/ and /k/ as voiced /b/ and /g/).

(4)

	/p/	/b/	/t/	/d/	/k/	/g/
Spanish	4	–138	9	–110	29	–108

The presumption in the literature on contemporary Celtic languages, heretofore, seems to be that a voiced plosive in Irish and Welsh, at least, could be devoiced in highly sonorous environments such as intervocalic position. Such a change has, indeed, been identified by Blust for the Austronesian languages Long Terawan (Blust 1992: 416) and Kiput (Blust 2002: 410–412; 2005: 241–246), but Beguš (under review), and Beguš & Nazarov (manuscript), argue, in effect, that such a change is a 'crazy rule' in the sense of Bach & Harms (1972; cf. Buckley 2000), a rule that is composed of several unattested stages such that, at the synchronic level, it does not bear any phonetic motivation.

As the VOT data in (3) and n. 7 make clear, the voiceless plosives are strongly aspirated in contemporary Celtic languages and the so-called voiced series of plosives is voiceless and lightly aspirated (i.e. they are characterised as unaspirated phonologically). The active Laryngeal feature in the plosive series is not [voice], but [spread glottis]. /p t k/, then, can be passively voiced to [p̌ t̬ k̬] in sonorous environments and voicing may occur as a phonetic enhancement (cf. Beckman et al. 2013). This voicing is gradient, i.e. the realisation of these plosives can range from completely voiceless to completely voiced.[8]

3. Evidence from older Celtic

There is some evidence from older stages of Celtic that the contrast in the plosive series was /p^h t^h k^h/ vs. /p t k/ and that such a contrast should be projected back into proto-Celtic. Just as contemporary Welsh indicates that labial and dorsal plosives after /s/ are unaspirated via orthographic ⟨sb⟩ and ⟨sg⟩ and Scottish Gaelic, at the dorsal place, via orthographic ⟨sg⟩, Middle Welsh, Early Modern Irish, and earlier Scottish Gaelic did likewise via the use of ⟨sb sd sg⟩ for [sp st sk];[9] cf. the following data from Middle Welsh, in which the initially cited forms exemplify attempts to represent unaspirated plosives after /s/ orthographically (cited after Schumacher 2011: 116):

(5) a. *kysbell* (PRBH 1182.37) 'appropriate, convenient' beside *kyspell* (LlH 210.1)
b. *kysdud* (CA 14.339) 'affliction, oppression' beside *kystut* (LlDC 51.30)
c. *kisgaw* (LlDC 13.17) 'I sleep' beside *kyscaf* (LlDC 28.62)

Even in the ancient Continental Celtic languages, there is some evidence that the contrast in the plosive series was /p^h t^h k^h/ vs. /p t k/:

(6) a. The Celtiberian acc. sg. det. **śDAḿ** (MLH K.6.1) occurs in one of a number of inscriptions in which an orthographic voicing contrast has been claimed to have been introduced (cf. Jordán Cólera 2005; 2007). But it does not make sense for a voiced plosive to occur immediately after /s/. It would make sense for ⟨**DA**⟩ to represent

[8] From the perspective of the listener, the release of the occlusion of plosives with a large VOT and the subsequent onset of voicing of the following vowel are perceived as two events, while the release of the occlusion of plosives with a small VOT and the subsequent onset of voicing of the following vowel may be 'perceived as occurring simultaneously, as in a [true] voiced plosive' (Klatt 1975: 695–696).

[9] As demonstrated by Kim (1970), in languages with a binary contrast in their plosive systems in which [spread glottis] is the active Laryngeal feature, groups of /s/ + voiceless plosive share a single [spread glottis] gesture which is extinguished prior to release of the plosive occlusion.

unaspirated [t] in this position, however, which implies that the ⟨T⟩ series of characters represents aspirated voiceless plosives (Eska, forthcoming).

b. It has been assumed that when the speakers of Cisalpine Celtic first adopted and adapted the north Etruscan alphabet – probably via the Veneti – their alphabet of Lugano, like the Venetic alphabets, employed the Etruscan voiceless unaspirated and voiceless aspirated plosive characters to indicate the contrast between voiceless and voiced plosives, respectively.[10] But the Celts subsequently gave up the aspirated characters. This hardly makes sense if Cisalpine Celtic contrasted /p t k/ vs. /b d g/. But it would make sense if the contrast was not one of [voice], but of [spread glottis], i.e. /p^h t^h k^h/ vs. /p t k/ (Eska 2017: 69–71).

c. Presumed /b d g/ in Celtic names embedded in Latin inscriptions which are otherwise entirely accurately spelt sometimes appear as ⟨P T C⟩, the very large majority at the dorsal place, where, owing to physiological factors, it is most difficult to maintain voicing (Eska, forthcoming):[11]

ATECNATI (AE 1974, 490; gen. sg.) beside ATE|GNATAE (CIL iii 5698; dat. sg.) < proto-Celt. *gnāto/ā- 'son/daughter'.

4. Laryngeal Realism and the fate of proto-IE */p/ (and */b/) in proto-Celtic

In this section, emphasis is placed upon providing phonetic reasons for sound changes and providing comparanda, rather than making simple stipulations.

(7) Stage I: Plosive system reconstructed for proto-Indo-European:

p t k q q^w
b d g ɢ $ɢ^w$
$b^ɦ$ $d^ɦ$ $g^ɦ$ $ɢ^ɦ$ $ɢ^{wɦ}$

The system of dorsal plosives normally reconstructed for proto-Indo-European contains a ternary contrast composed of palatal *$\hat{K}$, velar *K, and labial-velar *K^w. I here follow the system proposed by Michael Weiss apud Ringe 2017: 9^3. As related by Ringe, Weiss notes 'that shifts of palatal stops to velars are at best very rare in the attested historical phonologies of natural human languages. The palatals were also clearly the commonest dorsals (though not by a very wide margin), which suggests that they were typologically the unmarked set, i.e. probably really velars'; see further Huld (1997) and Kümmel (2007: 312–327). N.B. that virtually all languages that possess uvular plosives contrast them with velar plosives (Maddieson 2013b).

(8) Stage II: Pre-Celtic merger of uvulars with velars:

p t k k^w
b d g g^w
$b^ɦ$ $d^ɦ$ $g^ɦ$ $g^{wɦ}$

Cf. this development in central Palestinian Arabic dialects (Fischer & Jastrow 1980: 52); see further Kümmel (2007: 217–218).

[10] Though only at the coronal and dorsal places, presumably because proto-Celt. */k^w/ had not yet been labialised to /p/, so the *pi* character could unambiguously represent /b/.
[11] Epigraphic symbol: The pipe | indicates line breaks.

(9) Stage III: Proto-Celtic merger of */g^{w}/ with sparsely attested */b/ via labialisation:

p t k k^{w}
b d g
b^{ɦ} d^{ɦ} g^{ɦ} g^{wɦ}

Cf. this development in Middle Persian (Back 1978: 131) and Sardinian (Lausberg 1956: 24–35, 64–65); see further Kümmel (2007: 274).

(10) Stage IV: Proto-Celtic merger of voiced aspirated plosives with voiced unaspirated plosives:[12]

p t k k^{w}
b d g g^{w}

The proto-Indo-European aspirated voiced plosives were, in all likelihood, articulated as murmured [b̤ d̤ g̈], as in the modern Indic languages. The production of such a plosive is complex: should the glottal gesture be early or late or of insufficient intensity, a plain, unaspirated voiced plosive will be perceived (Stuart-Smith 2004: 170). Such a diachronic change is attested for proto-Anatolian, proto-Iranian, proto-Balto-Slavonic, and proto-Albanian, in addition to proto-Celtic. Such a change has also occurred in many contemporary northwestern Indic languages (Masica 1991: 102); see further Kümmel (2007: 95–96).

(11) Stage V: Proto-Celtic voiceless plosives become aspirated:

p^{h} t^{h} k^{h} k^{wh}
b d g g^{w}

Pedersen (1909: 242) states that the voiceless plosives of proto-Celtic were aspirated, but offers no opinion as to how they came to be so. Koch (1990: 198) argues that they developed as fortis allophones of lenis unaspirated voiceless plosives in Old Celtic. This is very possible at so-called strong positions, including word-initially (Keating et al. 1999; 2003; Scheer & Ségéral 2008: 135–136),[13] though I would date the change to proto-Celtic and allow that contact with (a) non-Indo-European substrate language(s) should not be ruled out as a contributing factor.[14] Such a development occurs in Baloči (Korn 2005: 222–225) and, as noted by Kümmel (2007: 169), must be reconstructed for proto-Germanic, for the shift of voiceless plosives to fricatives by Grimm's Law has no phonetic rationale in strong positions unless the plosives were aspirated. On the addition of aspiration to the articulation of voiceless plosives, see further Kümmel (2007: 168–170).

(12) Stage VI: Proto-Celtic voiced plosives become unaspirated voiceless plosives:

p^{h} t^{h} k^{h} k^{wh}
p t k k^{w}

The devoicing of voiced plosives is often triggered by the previous aspiration of the voiceless plosives (Kümmel 2007: 138) via a pull chain shift of the well-known type, for example, as in Grimm's Law in Germanic, of which this precise change forms a part (Iverson & Salmons 2003: 53–59).

[12] McCone (1996: 43) suggests that this change may have been triggered by the absence of an aspirated voiceless plosive series or led by */g^{wɦ}/ filling in the gap created by previous proto-IE */g^{w}/ > proto-Celt. */b/.

[13] In some languages, though not, at least, in Insular Celtic, syllable-initial is also a strong position.

[14] Iverson & Salmons (2003: 55) raise the same possibility for the rise of aspiration in voiceless plosives in Germanic.

(13) Stage VII: Proto-Celt. */p^h/ > */ɸ/ (earliest Cisalpine Celtic):
ɸ t^h k^h k^{wh}
p t k k^w

The frication of voiceless labial plosives, whether aspirated or unaspirated, is frequently attested (Kümmel 2007: 57). Cf. the tendency for Bangla[15] and Nepali /p^h/ > [ɸ]; so also only in initial position in Garhwali (Masica 1991: 103). This frication is also common in Samoyedic languages (Mikola 1998: 229).

It is likely that this stage is attested in earliest Cisalpine Celtic. The Alphabet of Lugano, in which most of the corpus is engraved, does not (usually) allow us to distinguish the plosive series, but /ɸ/ < proto-IE */p/, represented by the character ⟨**v**⟩, is attested in the form **uvamoKozis** (CIS 65 = CIM 180 = LexLep CO–48) < **upamo*- < *h_x*upm̥mo*- 'highest' (Eska 1998: 68–74; 2013).

(14) State VIII: Developments of proto-Celt. */ɸ/:
a. *ɸ > h > ∅ / {#__; V__V}

Debuccalisation of */ɸ/ occurs owing to the non-perception of turbulence because of the lack of downstream resonating chamber (Ohala & Solé 2010: 83), then loss of */h/. Cf. Cl. Arm. */ɸ/ > /h/ (sporadically > ∅) / {#__; V__V} (Meillet 1936: 30–31, 38–39) and Gk. /h/ > ∅ / {#__; V__V} (Lejeune 1972: 279–282); see further Kümmel (2007: 104, 116–119).

b. *ɸ > β / V__L

*/ɸ/ is passively voiced to [β] in the highly sonorous environment between a vowel and a liquid.

c. *ɸ > *β > w / {a; o}__n

After */ɸ/ is passively voiced to [β] in the highly sonorous environment between a vowel and /n/, the fricative is glided to /w/ before /n/ owing to the anticipatory lowering of the velum and the resulting attenuation of turbulence (Solé 2007). Cf. Lat. *ligna* 'line' > dialectal S. Ital. [ˈlɛwna] (Recasens 2002: 338).

d. *ɸ > x / __{s, t}[16]

*/ɸ/ > /x/ occurs owing to labials and velars having an important acoustic feature, viz. low F_2, in common (Ohala & Lorentz 1977: 581–582); cf. Span. Cat. Port. */ɸ/ > /x/ / __s (Lausberg 1956: 50) and Umb. /f/ > */x/ > /h/ / __t (Meiser 1986: 178–179); see further Kümmel (2007: 222).

Note that proto-IE */sp/ is continued unchanged because the two segments share a single [spread glottis] gesture which is extinguished prior to the release of the plosive occlusion (Kim 1970), i.e. though the plosive in this group continues proto-IE */p/, it was not phonetically realised as early proto-Celt. *[p^h] and hence did not fricate to */ɸ/; cf. Latinised Transalp. Celt. SPIR|NVS (CIL xiii 11558) and *Bratuspantium* (Caesar, *BG* ii 13).[17] Cf. also Germanic, in which the plosive in */sp/ groups does not fricate to */f/ via Grimm's Law because it was not

[15] Formerly known as Bengali.

[16] This change presumably occurs after */k/ > /x/ / __{s, t}, which took place because the [spread glottis] gesture of the inherited plosive was not realised as aspiration before an obstruent, but, instead, resulted in frication (Iverson & Salmons 1995: 387–388); see Vaux (1998) on the [spread glottis] specification of fricatives.

[17] Proto-Celt. */sp/ later evolved to */sw/ in Goidelic owing to the labial being heard as /w/ because of similarity in formant transitions (Ohala & Lorentz 1977: 582); in Brittonic, it later evolved to */sɸ/ via spreading of [continuant] > */sf/ by partial place assimilation and then /f/ via dissimilatory loss.

aspirated, e.gg. Goth. acc. pl. *spilla* 'announcers', OIcel. *spjall*, OE *spell*, OS OHG *spel* 'speech, story' < proto-IE *(*s*)*pelh*$_x$- 'speak publicly'.

(15) Stage VIII^{bis} (Celtiberian, early Ogam Irish):

$$\begin{array}{llll} & t^h & k^h & k^{wh} \\ p & t & k & k^w \end{array}$$

In Celtiberian and earliest Goidelic, unlike early Cisalpine Celtic, proto-Celt. */ɸ/ has been completely lost in initial and intervocalic positions.

(16) Stage IX^a: Regional Celtic labial-velars merge with labial consonants (Transalpine Celtic and Brittonic):[18]

$$\begin{array}{lll} p^h & t^h & k^h \\ p & t & k \end{array}$$

*/k^{wh}/ > /p^h/ occurs owing to labials and velars having an important acoustic feature, viz. low F_2, in common (Ohala & Lorentz 1977: 581–582); see also Kümmel (2007: 274). */k^w/ > /w/ occurs owing to the labial-velar being heard as /w/ because of similarity in formant transitions (Ohala & Lorentz 1977: 582); cf. Transalp. Celt. *uediju=mj* 'I beseech' (RIG L–100) < proto-Celt. *k^w*et*- < *g^w*ed*- < proto-IE *$g^{wɦ}$*ed*^{ɦ}- 'request, wish'.

(17) Stage IX^b: Regional Celtic labial-velars merge with dorsals (later Ogam Irish):

$$\begin{array}{lll} & t^h & k^h \\ p & t & k \end{array}$$

As noted by Kümmel (2007: 275), a general delabialisation of obstruents is not uncommon; cf. Fren. /k^w g^w/ > /k g/ (Lausberg 1956: 22–25) and Alb. */k^w g^w/ > */k g/ (Demiraj 1997: 64).

5. Summary

The assumption that the plosive series of proto-Celtic were contrasted via the Laryngeal feature [spread glottis], rather than via [voice], not only accords with the phonetic facts of the contemporary Celtic languages, but makes sense of what have been considered to be orthographic anomalies in the Continental Celtic languages, and helps make sense of sound changes that must have occurred in proto-Celtic. In particular, the view that voiceless plosives became aspirated in proto-Celtic makes possible the change of the voiceless labial plosive to a fricative in word-initial position, a change that would not be phonetically motivated were the plosive unaspirated. All other necessary changes to arrive at the plosive systems of the various Celtic languages thereafter can be phonetically motivated without issue and are well paralleled cross-linguistically.

6. Coda. The loss of proto-IE */p/ in Celtic does not support the Celtic from the West hypothesis

A number of scholars have proposed that the loss of proto-IE */p/ in Celtic may be the result of prehistoric contact with non-Indo-European pre-Basque/Aquitanian or Iberian (McCone 1996: 43; 2005: 403; Ballester 2004: 114–117; 2012: 10–11; Jordán Cólera 2004: 68; Schrijver 2015: 200); those who advocate for the Celtic from the West hypothesis – which places the proto-Celtic homeland in the Iberian Peninsula – are virtually certain that such contact is

[18] Some relic forms of /k^{wh}/ are preserved in Transalpine Celtic, e.g. the nom. pl. ethnonym A|RESE|QVANI (RIG *L–12).

responsible for the loss of Celt. */p/ (Koch 2010: 294–295; 2011: 171; 2013a: 123; 2013b: 264–265; 2014: 8; 2016: 468; Vennemann 2016: 511–527).

But this view ignores the fact that there are numerous languages which lack /p/ (Maddieson 2013a).[19] This is because the release of the occlusion of /p/ often is not perceptually salient owing to the lack of a downstream resonating chamber (Ohala & Solé 2010: 72–73).

More importantly for the matter at hand, however, are the facts that /p/, though rare, was not entirely absent from pre-Basque and Aquitanian (Michelena 1957: 130) and that contemporary Basq. /p/ is phonologically unaspirated. In pre-Basque, in fact, plosives possessed only a fortis-lenis contrast, which was reïnterpreted as a voicing contrast at a later date (Trask 1997: 125). Most northern dialects of contemporary Basque do possess phonetically aspirated voiceless plosives beside unaspirated voiceless plosives, but their etymological origin is identical and they, in most dialects, do not contrast.[20] This aspiration, in pre-Basque, as well as in contemporary Basque dialects, was suprasegmental and non-contrastive (Trask 1997: 158, 168), and seems originally to have been associated with the usual placement of the accent on the second syllable in forms of two or more syllables in pre-Basque (Trask 1997: 127; Igartua Ugarte 2001).[21] In ancient Aquitanian, which seems securely to be closely related to pre-Basque (see Trask 1997: 398–403), the placement of aspiration, which is robustly attested, is virtually identical to that in contemporary Basque dialects (Trask 1997: 401).

Iberian inscriptions engraved in Greek characters reveal that the language lacked /p/, but those inscriptions appear to demonstrate that /t k/ contrasted with /d g/ (Untermann 1990: 155); the complete absence of orthographic ⟨φ θ χ⟩ (Untermann 1990: 133), furthermore, suggests that [spread glottis], i.e. aspiration, was phonologically inert in Iberian.

The pre-Basque/Aquitanian and Iberian facts are important because it was proto-Celt. */p^h/ < proto-IE */p/ that was lost, not proto-Celt. */p/ < proto-IE */b/ (including earlier proto-Celt. */b/ < proto-IE */g^w/), which remains unaltered to the present day. Had there been a contact effect of pre-Basque/Aquitanian or Iberian upon the Celtic labial plosives, it would have been the latter which was affected, which is not what is in evidence.[22] Furthermore, as described supra, it is clear that proto-Celt. */p^h/ was not lost outright, but was initially fricated to */ɸ/ in all environments; no labial fricative, however, is reconstructed for pre-Basque (Trask 1997: 126) or Iberian (Untermann 1990: 152–155). We can be confident that this pillar of the Celtic from the West hypothesis is disproved.

Abbreviations

AE	= *L'année épigraphique*
CA	= Williams (1961)

[19] Maddieson's language sample has most tokens of languages lacking /p/ concentrated in Africa north of the equator with an outlying area in New Guinea, leading him to question whether there are general phonetic principles behind the absence of /p/ and suggest that areal factors are responsible. In my view, his view is erroneous and due to a sampling error. His map, for example, does not include ancient languages, so early Old Irish and Iberian, for example, are not present.

[20] Trask (1997: 162) notes that 'the functional load of the aspiration is not great, particularly on voiceless plosives', i.e. in dialects that possess aspirated voiceless plosives, they are usually in free variation with unaspirated voiceless plosives.

[21] Though the accent could also fall upon the first syllable. For a conspectus upon the views about aspiration in early Basque, see Salaberri Zaratiegi & Salaberri Izko (2016). For the complex rules of aspiration location in contemporary Basque, see Trask (1997: 157–163).

[22] We can also allow that, since neither pre-Basque/Aquitanian nor Iberian possessed phonologically aspirated plosives, they could have influenced both proto-Celt. /p^h/ and /p/. But this clearly did not happen, for, as noted supra, Celt. /p/ < earlier */b/ is preserved to the present day.

CIL	=	*Corpus inscriptionum Latinarum*
CIM	=	Morandi (2004)
CIS	=	Solinas (1995)
LexLep	=	*Lexicon Leponticum* ⟨http://www.univie.ac.at/lexlep/wiki/Main_Page⟩
LlDC	=	Jarman (1982)
LlH	=	Morris-Jones & Parry-Williams (1971)
MLH K	=	Untermann (1997: 349–722)
PRBH	=	Evans (1911–1926)
RIG L–1–*16	=	Lejeune (1988: 55–194)
RIG L–18–*139	=	Lambert (2002)

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